Assignment 3

1. Global methane levels from 2002

Methane (CH_4) is a naturally occurring greenhouse gas, but one whose abundance has been increased substantially above its pre-industrial value by human activities, primarily agricultural emissions (rice production, ruminants) and fossil fuel production and use. A clear annual cycle is largely due to seasonal wetland emissions. Satellite instruments do not measure methane directly: they measure spectrally resolved near-infrared and infrared radiation, and the averaged methane abundance in the sampled atmospheric column is retrieved from the absorption lines in those radiance observations. In this problem set the merged product gives monthly averaged methane (*xch4*) in ppb on a 5° (lon) × 5° (lat) grid over the globe from 2003-01 to 2020-06.

1.1 [5 points] Compute methane climatology for each month, and plot your results in 12 panels.

I open the file first and check the unit. The variable is stored in mole/mole, so I multiply it by 10^9 to work in ppb, which is the unit asked for in the problem.

In [2]:

```
# Open the methane file
ch4 = xr.open_dataset("200301_202006-C3S-L3_GHG-PRODUCTS-OBS4MIPS-MERGED-v4.3.nc",
engine="netcdf4")

# Missing retrievals are written as 1e20 in this file and flagged by the missing_value attribute.
Mask them out by hand as well, so the arithmetic does not depend on the attribute being decoded.
xch4 = ch4["xch4"].where(ch4["xch4"] < 1e-3)

# xch4 is stored in mole/mole in this file, convert it to ppb
xch4 = xch4 * 1e9
xch4.attrs["units"] = "ppb"

# Check the dimensions and the unit
print(xch4.dims, xch4.shape)
print("units:", xch4.attrs["units"])
```

```
('time', 'lat', 'lon') (210, 36, 72)
units: ppb
```

The climatology is the average of all Januaries, all Februaries and so on, which is `groupby("time.month").mean("time")` in xarray. About 72 % of the values are missing (failed retrievals over cloud, bright surfaces and the polar night), and January 2015 has no data at all, so before averaging I mask the cells that have fewer than 6 valid months. Otherwise a cell with one or two values would carry as much weight as a cell with 17 years of data.

In [3]:

```
# Climatology of each calendar month, i.e. average all Januaries, Februaries, ...
xch4_clim = xch4.groupby("time.month").mean(dim="time")

# Grid cells with less than 6 months of data are too noisy to average, mask them out
xch4_clim = xch4_clim.where(xch4.notnull().sum(dim="time") >= 6)

# Month names for the panel titles
xch4_clim = xch4_clim.assign_coords(month=month_names)

# One map per month
print(xch4_clim.shape)
```

```
(12, 36, 72)
```

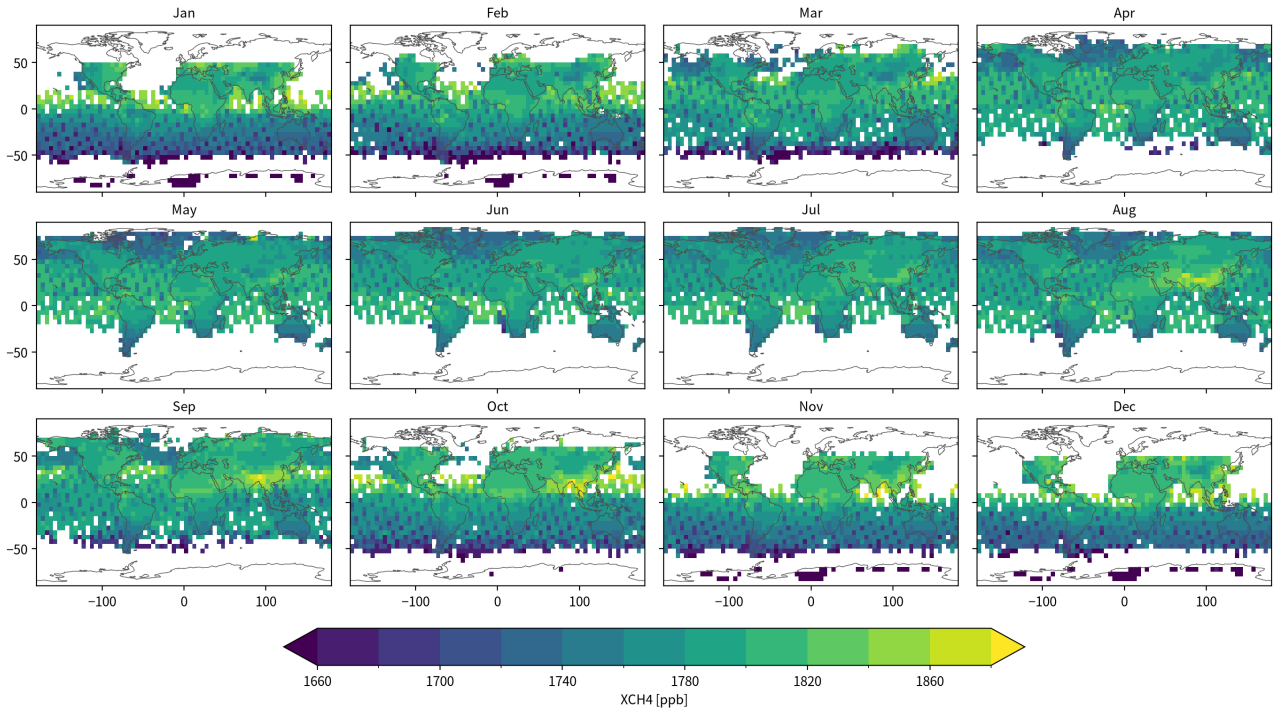
All 12 panels share one color scale, so I plot them through the facet interface of xarray. cartopy does not work on my laptop, so the maps use plain longitude-latitude axes instead of a map projection, and the coastlines are drawn by hand from a separate Natural Earth file (coastlines.npz) that is kept next to the data.

In [4]:

```
# 12 panels, one per month, all with the same color scale
fg = xch4_clim.plot(col="month", col_wrap=4, figsize=(14, 8.5),
                    levels=np.arange(1660, 1881, 20), cmap="viridis", extend="both",
                    cbar_kwargs={"orientation": "horizontal", "shrink": 0.6,
                                 "pad": 0.06, "label": "XCH4 [ppb]"})
fg.set_titles("{value}")

# Coastlines for every panel, the axis labels are only kept on the outer edges
for ax in fg.axs.flat:
    draw_coast(ax)
    ax.set_xlabel("")
    ax.set_ylabel("")

fg.fig.savefig("figs/fig1_1.png", dpi=200, bbox_inches="tight")
plt.show()
```



The climatology is about 1810-1850 ppb over the northern mid-latitudes but only 1660-1720 ppb over the southern high latitudes. Nearly all methane sources (wetlands, rice fields, ruminants, fossil fuel use) are on land north of the equator, and the inter-hemispheric exchange time is about one year, so the gradient survives. The seasonal cycle is clearest over northern land, where *xch4* peaks in July-September when wetland emission is strongest, and is weakest over the southern ocean, where the seasonal change of sources is small. White cells are the masked or missing retrievals; the rows next to the poles are empty because no retrieval exists there.

1.2 [5 points] Plot globally-averaged methane from 2003-01 to 2020-06 as a time series.

A global mean is not a plain average over grid cells, because a 5° cell covers less area near the poles. I weight every cell by cos(lat) and let xarray skip the missing cells, then fit a straight line to the monthly series to get the trend.

In [5]:

```
# Area of a 5x5 grid cell is proportional to cos(lat), so use cos(lat) as weight
weights = np.cos(np.deg2rad(xch4.lat))

# Global mean, cells with missing data are skipped
xch4_global = xch4.weighted(weights).mean(dim=["lat", "lon"])

# Linear trend over the whole period
t = (xch4_global.time.dt.year + (xch4_global.time.dt.month - 1) / 12).where(xch4_global.notnull())
slope = np.polyfit(t.dropna("time"), xch4_global.dropna("time"), 1)[0]
print(f"Linear trend: {slope:.2f} ppb/yr")
```

```
Linear trend: 6.44 ppb/yr
```

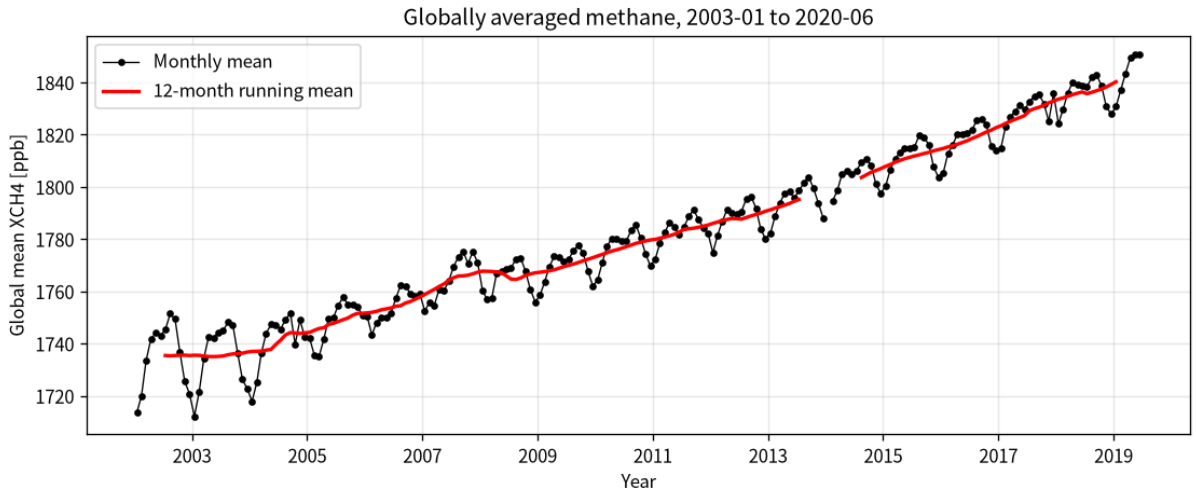
In [6]:

```
fig, ax = plt.subplots(figsize=(11, 4), dpi=120)

# Monthly values and the 12-month running mean
ax.plot(xch4_global.time, xch4_global, "o-", color="k", markersize=3, linewidth=0.8,
        label="Monthly mean")
ax.plot(xch4_global.time, xch4_global.rolling(time=12, center=True).mean(),
        color="r", linewidth=2, label="12-month running mean")

ax.set(xlabel="Year", ylabel="Global mean XCH4 [ppb]",
       title="Globally averaged methane, 2003-01 to 2020-06")
ax.grid(alpha=0.3)
ax.legend()

plt.savefig("figs/fig1_2.png", bbox_inches="tight")
plt.show()
```



The global mean rises from 1713 ppb in January 2003 to 1851 ppb in June 2020, about 138 ppb in total, and the linear trend is 6.44 ppb/yr. The growth is not steady: 2003-2006 is almost flat, the curve climbs again from 2007, and the increase accelerates after 2014. The shape agrees with the NASA SVS animation (svs.gsfc.nasa.gov/5007); the absolute values are a few tens of ppb higher than that curve, which is a difference between retrieval products rather than a problem of this time series.

The annual cycle of about 25 ppb is superimposed on the trend: 1761.6 ppb in January (lowest), 1787.3 ppb in April-May (highest) and a second peak in August. The phase follows the wetland emission maximum in summer and the weaker OH sink in winter. A few months are missing in the raw product, and January 2015 is a visible gap in the series.

1.3 [5 points] Plot deseasonalized methane levels at point [15°S, 150°W] from 2003-01 to 2020-06 as a time series.

I take the grid point nearest to 15°S, 150°W, which is (-12.5°, -147.5°), and look at how complete it is. Persistent cloud over the warm pool means 133 of the 210 months have no retrieval, so I fill the gaps by linear interpolation in time before removing the seasonal cycle.

In [7]:

```
# Grid point closest to [15S, 150W]; longitude in this file runs from -180 to 180
xch4_pt = xch4.sel(lat=-15, lon=-150, method="nearest")
print("grid point:", float(xch4_pt.lat), float(xch4_pt.lon))
print("missing months:", int(xch4_pt.isnull().sum()))

# Clouds and retrieval failures leave gaps, fill them by linear interpolation in time
xch4_pt = xch4_pt.interpolate_na(dim="time", method="linear")

# Subtract the climatology of the same calendar month to remove the seasonal cycle
xch4_pt_anom = xch4_pt.groupby("time.month") - xch4_pt.groupby("time.month").mean(dim="time")
```

```
grid point: -12.5 -147.5
missing months: 133
```

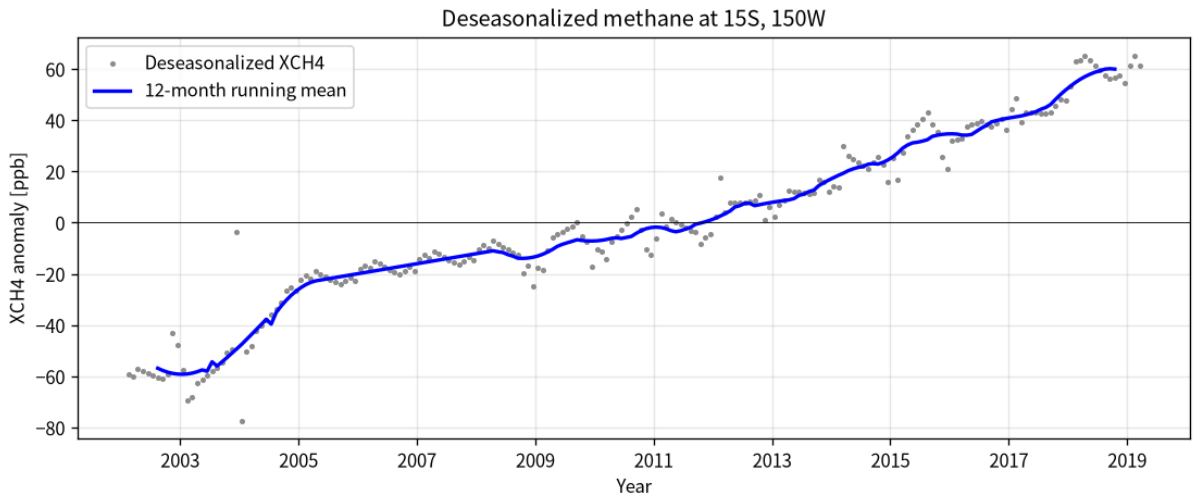
In [8]:

```
fig, ax = plt.subplots(figsize=(11, 4), dpi=120)

ax.plot(xch4_pt_anom.time, xch4_pt_anom, ".", color="0.55", markersize=4,
        label="Deseasonalized XCH4")
ax.plot(xch4_pt_anom.time, xch4_pt_anom.rolling(time=12, center=True).mean(),
        color="b", linewidth=2, label="12-month running mean")
ax.axhline(0, color="k", linewidth=0.5)

ax.set(xlabel="Year", ylabel="XCH4 anomaly [ppb]",
       title="Deseasonalized methane at 15S, 150W")
ax.grid(alpha=0.3)
ax.legend()

plt.savefig("figs/fig1_3.png", bbox_inches="tight")
plt.show()
```



At this point the deseasonalized anomaly climbs from about -60 ppb in 2003 to +60 ppb in 2019, the same 110-120 ppb increase as the global mean, with a slope of about 6.5 ppb/yr. The 12-month running mean is flat in 2009-2011 and rises more slowly around 2013-2014. The remaining scatter of about ±15 ppb is retrieval noise, and since two thirds of the months are interpolated the short-term wiggles should not be read as real variations.

2. Niño 3.4 index

The Niño 3.4 anomalies may be thought of as representing the average equatorial sea surface temperature across the Pacific from about the dateline to the South American coast (5N-5S, 170W-120W). The index typically uses a 3-month running mean, and El Niño or La Niña events are defined when the anomaly exceeds ±0.5 °C for 5 months or longer. The data used here is NOAA_NCDC_ERSST_v3b_SST.nc, monthly SST from 1960-01 to 2016-12 on a 2° grid.

2.1 [10 points] Compute monthly climatology for SST from the Niño 3.4 region, and subtract climatology from the SST time series to obtain anomalies.

The longitude in this file runs from 0 to 360, so 170W is 190E and 120W is 240E. Writing `slice(-170, -120)` would select an empty box, which is the easy trap here.

In [9]:

```
# Open the SST file
sst_ds = xr.open_dataset("NOAA_NCDC_ERSST_v3b_SST.nc", engine="netcdf4")

# Land points are filled with -999, mask them even if the attribute is not decoded
sst = sst_ds["sst"].where(sst_ds["sst"] > -100)

# Niño 3.4 region; longitude here runs 0-360, so 170W is 190E and 120W is 240E
nino34 = sst.sel(lat=slice(-5, 5), lon=slice(190, 240))
nino34
```

```
<xarray.DataArray 'sst' (time: 684, lat: 5, lon: 26)> [88920 values with dtype: float32]
Coordinates:
  * time (time) datetime64[ns] 1960-01-15 ... 2016-12-15
  * lat (lat) float32 -4.0 -2.0 0.0 2.0 4.0
  * lon (lon) float32 190.0 192.0 ... 238.0 240.0
```

Then the climatology is the mean of each calendar month, the anomaly is the monthly value minus that climatology, and the index is the average of the anomaly over the box followed by a 3-month running mean.

In [10]:

```
# Monthly climatology of the Niño 3.4 SST
nino34_clim = nino34.groupby("time.month").mean(dim="time")

# Anomaly = SST minus the climatology of the same calendar month
nino34_anom = nino34.groupby("time.month") - nino34_clim

# Average over the box to get the index
nino34_index = nino34_anom.mean(dim=["lat", "lon"])

# 3-month running mean, as used for the operational ONI
nino34_index_3m = nino34_index.rolling(time=3, center=True).mean()

print("box mean climatology:", float(nino34_clim.mean()))
```

```
box mean climatology: 27.035411834716797
```

I plot the region on a map together with the index itself, so that the box and the time series can be compared at a glance.

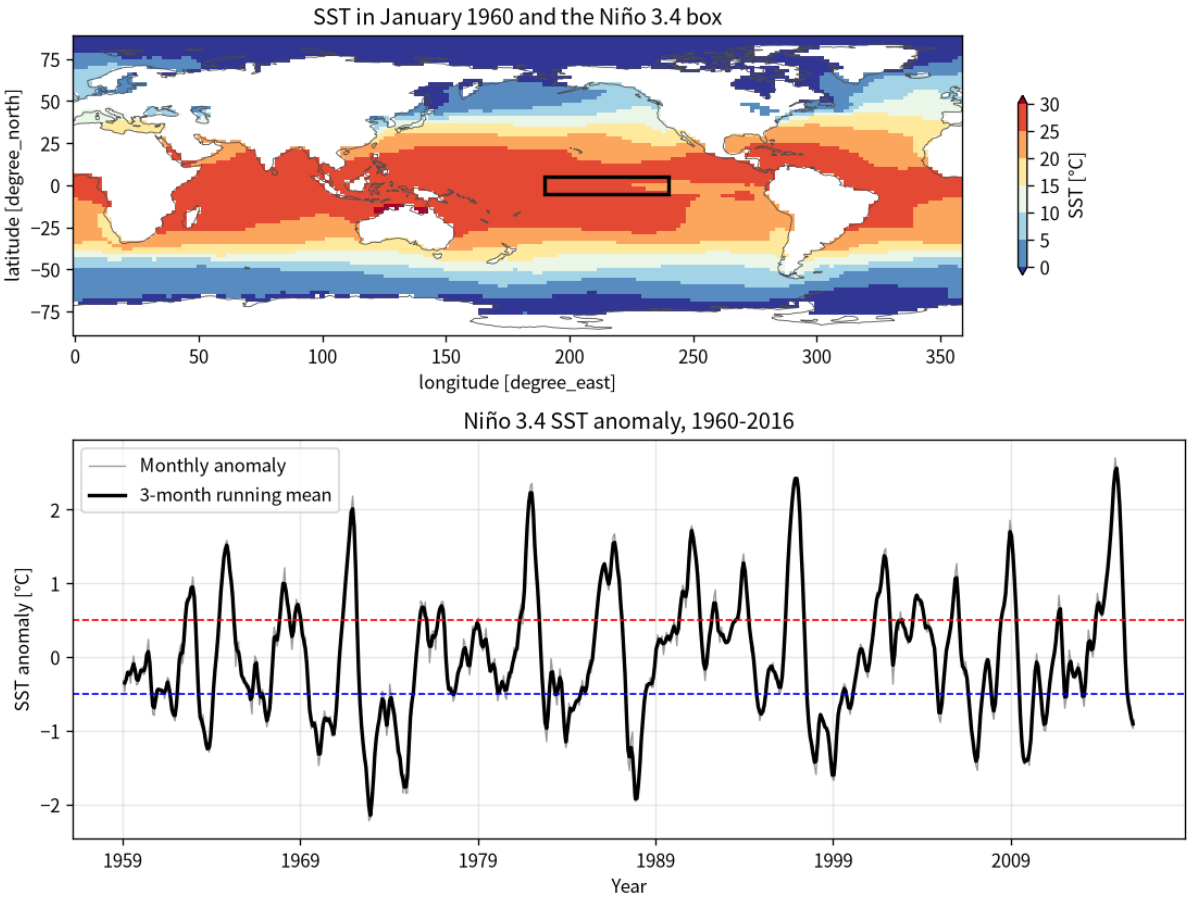
In [11]:

```
fig = plt.figure(figsize=(11, 8), dpi=120)
grid = plt.GridSpec(2, 1, height_ratios=[1.2, 1.6], hspace=0.3)

# Map of the region
ax = fig.add_subplot(grid[0])
sst.isel(time=0).plot(ax=ax, cmap="RdYlBu_r",
                      levels=np.arange(0, 31, 5), extend="both",
                      cbar_kwargs={"label": "SST [°C]", "shrink": 0.6})
draw_coast(ax)
ax.add_patch(plt.Rectangle((190, -5), 50, 10, fill=False, edgecolor="k",
                           linewidth=2))
ax.set_title("SST in January 1960 and the Niño 3.4 box")

# Index
ax = fig.add_subplot(grid[1])
ax.plot(nino34_index.time, nino34_index, color="0.6", linewidth=0.8,
        label="Monthly anomaly")
ax.plot(nino34_index_3m.time, nino34_index_3m, color="k", linewidth=2,
        label="3-month running mean")
ax.axhline(0.5, color="r", linestyle="--", linewidth=1)
ax.axhline(-0.5, color="b", linestyle="--", linewidth=1)
ax.set(xlabel="Year", ylabel="SST anomaly [°C]",
       title="Niño 3.4 SST anomaly, 1960-2016")
ax.grid(alpha=0.3)
ax.legend(loc="upper left")

plt.savefig("figs/fig2_1.png", bbox_inches="tight")
plt.show()
```



The climatology of the box is 27.0 °C. The anomalies run between -2.2 and +2.7 °C: the strongest warm anomaly is +2.70 °C at the end of 1997 (the 1997-98 El Niño), the strongest cold anomaly is -2.22 °C in 1973-74. After the 3-month running mean the peaks are +2.56 °C (1997-98) and +2.38 °C (2015-16), the troughs -2.15 °C (1973-74). Warm and cold phases each last one to two years and alternate irregularly with a period of about 3-7 years. Over 1960-2016 there are 175 months above +0.5 °C and 196 months below -0.5 °C.

2.2 [10 points] Visualize the computed Niño 3.4. Your plot should look similar to the NOAA monitoring figure.

I copy the layout of the NOAA figure: one bar per month, red above +0.5 °C, blue below -0.5 °C and grey in between, a black 3-month running mean on top, the two thresholds as dotted lines, and one tick every four years. The bars are drawn against their index number rather than the date axis, because bars have a fixed width while months do not.

In [12]:

```
fig, ax = plt.subplots(figsize=(12, 4.5), dpi=120)

# Bars are red for El Niño (>0.5°C), blue for La Niña (<-0.5°C), grey in between
colors = np.where(nino34_index >= 0.5, "r", np.where(nino34_index <= -0.5, "b", "0.8"))
ax.bar(np.arange(nino34_index.size), nino34_index, color=colors, width=0.8)

# 3-month running mean
ax.plot(np.arange(nino34_index.size), nino34_index_3m, color="k", linewidth=1.6,
        label="3-month running mean")

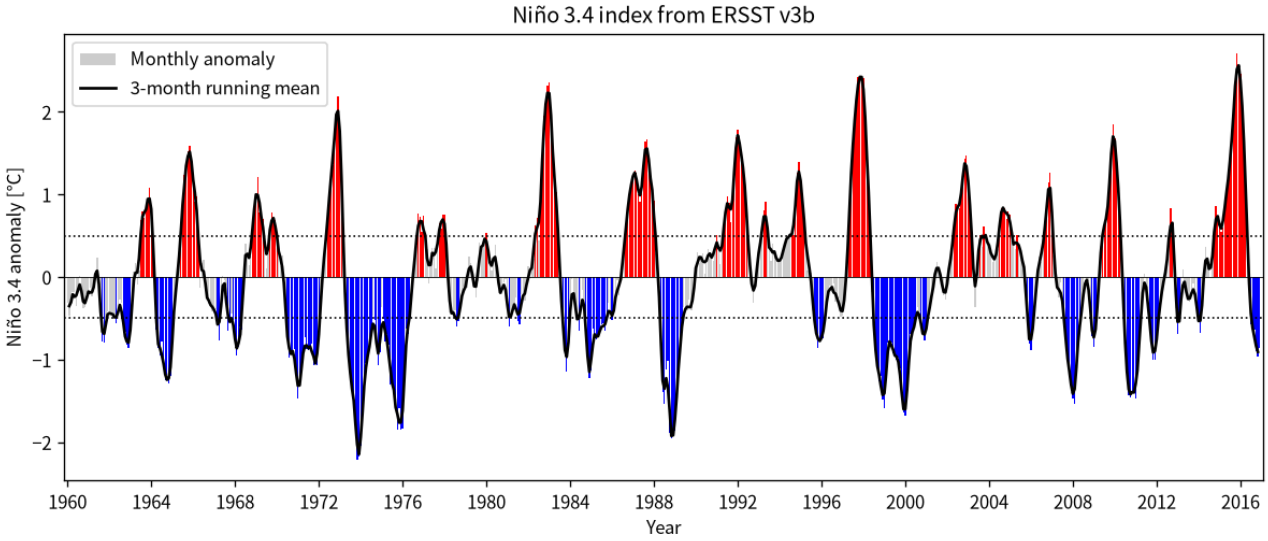
# Thresholds
ax.axhline(0.5, color="k", linestyle=":", linewidth=1)
ax.axhline(-0.5, color="k", linestyle=":", linewidth=1)
ax.axhline(0, color="k", linewidth=0.5)

# One tick every 4 years, at January
years = nino34_index.time.dt.year.values
months = nino34_index.time.dt.month.values
ticks = [i for i in range(nino34_index.size) if months[i] == 1 and years[i] % 4 == 0]
ax.set_xticks(ticks)
ax.set_xticklabels(years[ticks])

ax.set(xlabel="Year", ylabel="Niño 3.4 anomaly [°C]",
       title="Niño 3.4 index from ERSST v3b")

# Legend with proxy handles, the bars are colored one by one
handles = [Patch(facecolor="0.8", label="Monthly anomaly"),
           Line2D([], [], color="k", linewidth=1.6, label="3-month running mean")]
ax.legend(handles=handles, loc="upper left")
ax.set_xlim(-2, nino34_index.size + 1)

plt.savefig("figs/fig2_2.png", bbox_inches="tight")
plt.show()
```



The shape matches the NOAA monitoring plot. The warm events are 1963-64, 1965-66, 1968-69, 1972-73, 1976-77, 1982-83, 1987-88, 1991-92, 1994-95, 1997-98, 2002-03, 2004-05, 2006-07, 2009-10 and 2014-16; the cold events are 1964-65, 1970-72, 1973-74, 1975-76, 1988-89, 1998-99, 1999-2000, 2007-08 and 2010-12. 1997-98 and 2015-16 are the two strongest warm events, 1973-74 and 1988-89 the two strongest cold events. The running mean is a little smoother and weaker than the monthly peaks, which is expected from a 3-month filter.

3. Explore a netCDF dataset

I browsed GES DISC for a dataset with a time dimension and chose GPCP monthly precipitation version 2.3, produced by NASA GSFC and distributed by GES DISC as GPCPMON. The file precip.mon.mean.nc, about 19 MB, had been downloaded once from the NOAA PSL mirror of the same product and is now kept in the notebook folder, so this part of the notebook only reads a local file and needs no internet access. The data is a 2.5° global grid from 1979 to the present, with the precipitation rate in mm/day.

One detail about this file: its auxiliary variable *time_bnds* carries fill values and an irregular unit string (*days since 1800-01-01 00:00:00 0:00*), which makes the automatic time decoding fail with ValueError/ OverflowError. Dropping that variable when opening the file solves it.

In [13]:

```
# Open the local GPCP file, no download needed time_bnds contains fill values that break the
automatic time decoding, so it is dropped
precip_ds = xr.open_dataset("precip.mon.mean.nc", engine="netcdf4", drop_variables=["time_bnds"])

# Missing cells are filled with -9.96921e36, mask them by hand as well
precip = precip_ds["precip"].where(precip_ds["precip"] >= 0)
precip
```

```
<xarray.DataArray 'precip' (time: 570, lat: 72, lon: 144)> 24MB
Coordinates:
  * time (time) datetime64[ns] 1979-01-01 ... 2026-06-01
  * lat (lat) float32 -88.75 -86.25 ... 86.25 88.75
  * lon (lon) float32 1.25 3.75 ... 356.25 358.75
```

3.1 [5 points] Plot a time series of a certain variable with monthly seasonal cycle removed.

Same recipe as in 1.2: area weighted global mean, then subtract the climatology of each calendar month.

In [14]:

```
# Area weighting and the global mean
weights = np.cos(np.deg2rad(precip.lat))
precip_global = precip.weighted(weights).mean(dim=["lat", "lon"])

# Remove the monthly climatology
precip_clim = precip_global.groupby("time.month").mean(dim="time")
precip_anom = precip_global.groupby("time.month") - precip_clim

print("global mean:", float(precip_global.mean()), "mm/day")
print("anomaly range:", float(precip_anom.min()), float(precip_anom.max()))
```

```
global mean: 2.687621593475342 mm/day
anomaly range: -0.1788170337677002 0.1288747787475586
```

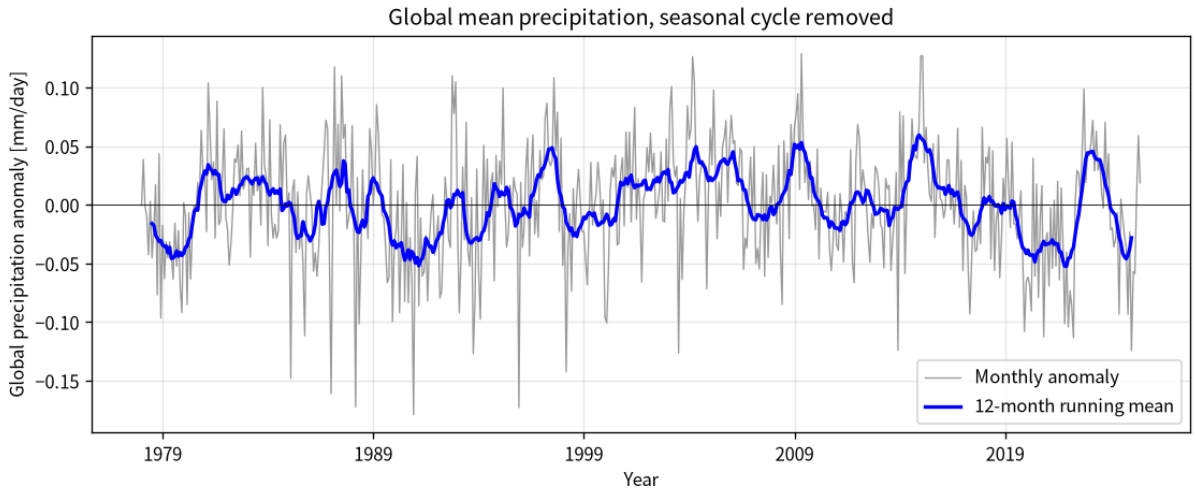
In [15]:

```
fig, ax = plt.subplots(figsize=(11, 4), dpi=120)

ax.plot(precip_anom.time, precip_anom, color="0.6", linewidth=0.8,
        label="Monthly anomaly")
ax.plot(precip_anom.time, precip_anom.rolling(time=12, center=True).mean(),
        color="b", linewidth=2, label="12-month running mean")
ax.axhline(0, color="k", linewidth=0.5)

ax.set(xlabel="Year", ylabel="Global precipitation anomaly [mm/day]",
       title="Global mean precipitation, seasonal cycle removed")
ax.grid(alpha=0.3)
ax.legend()

plt.savefig("figs/fig3_1.png", bbox_inches="tight")
plt.show()
```



The global mean is 2.69 mm/day and its seasonal cycle is weak, the climatology stays between 2.63 and 2.72 mm/day, because the rain belt only moves from one hemisphere to the other. The anomalies are ±0.18 mm/day, but the El Niño years stand out: 1982-83, 1987-88, 1991-92, 1997-98 and 2015-16 are positive, since the warm-pool rain moves into the central Pacific, while the two years after the Pinatubo eruption (1991-1993) are negative. There is no clear global trend over these 47 years.

3.2 [10 points] Make at least 5 different plots using the dataset.

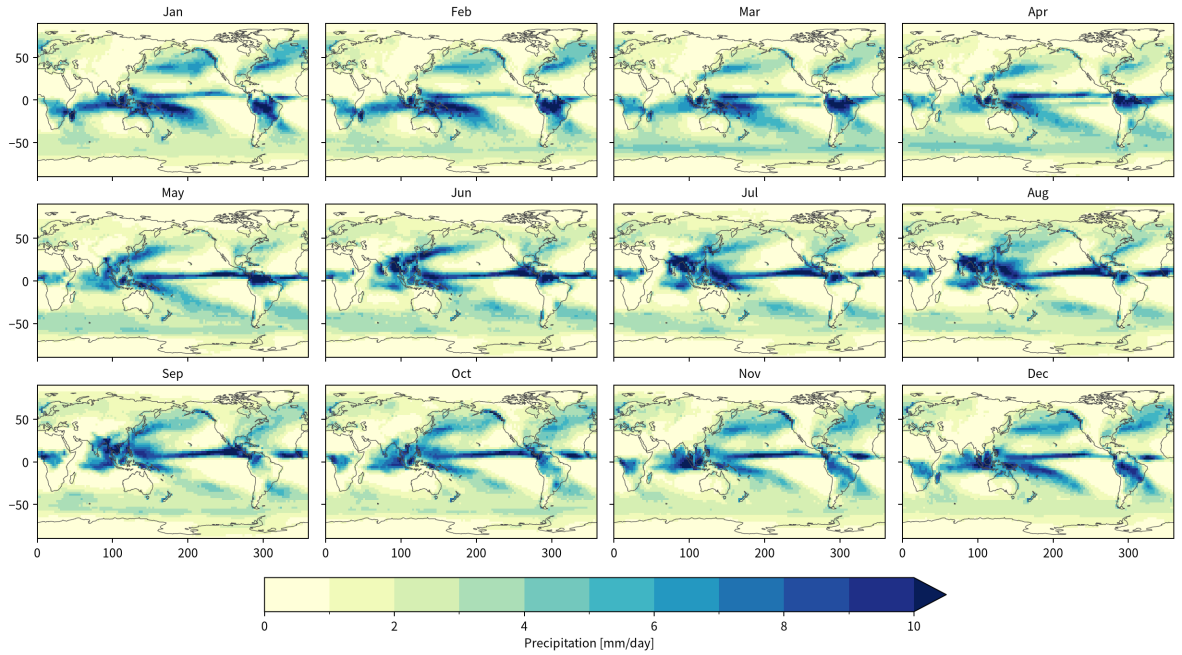
(1) The monthly climatology, in the same 12-panel form as problem 1.

In [16]:

```
# (1) Monthly climatology of precipitation
precip_month_clim = precip.groupby("time.month").mean(dim="time")
precip_month_clim = precip_month_clim.assign_coords(month=month_names)

fg = precip_month_clim.plot(col="month", col_wrap=4, figsize=(14, 8.5),
                            levels=np.arange(0, 10.1, 1), cmap="YlGnBu", extend="max",
                            cbar_kwargs={"orientation": "horizontal", "shrink": 0.6,
                                         "pad": 0.06, "label": "Precipitation [mm/day]"})
fg.set_titles("{value}")
for ax in fg.axs.flat:
    draw_coast(ax)
    ax.set_xlabel("")
    ax.set_ylabel("")

fg.fig.savefig("figs/fig3_2a.png", dpi=180, bbox_inches="tight")
plt.show()
```



The panels show the ITCZ moving from south of the equator in January-March to about 10°N in July-September, the Indian and East Asian monsoon rain from June to September, the Amazon wet season from December to April, and the dry belt over North Africa.

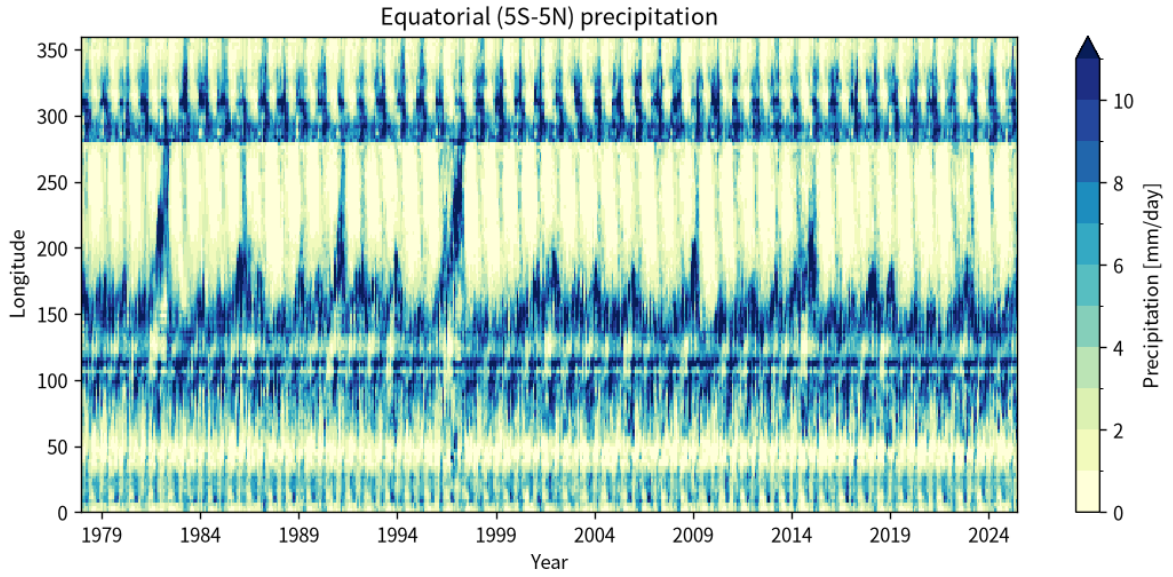
(2) A Hovmöller diagram of the equatorial band, longitude against time.

In [17]:

```
# (2) Hovmoeller diagram: longitude against time, averaged over 5S-5N
fig, ax = plt.subplots(figsize=(11, 4.5), dpi=120)

precip.sel(lat=slice(-5, 5)).mean(dim="lat").plot(ax=ax, x="time", y="lon",
                                                  cmap="YlGnBu",
                                                  levels=np.arange(0, 12, 1), extend="max",
                                                  cbar_kwargs={"label": "Precipitation [mm/day]"})
ax.set(xlabel="Year", ylabel="Longitude", title="Equatorial (5S-5N) precipitation")

plt.savefig("figs/fig3_2b.png", bbox_inches="tight")
plt.show()
```



Two activity centres persist near 110-150°E and 160-200°E. During the 1997-98 El Niño the rain moves east to 220-250°E and stays there for more than a year, and 2015-16 shows a weaker version of the same shift. The vertical stripes are the annual cycle repeating at each longitude.

(3) The linear trend of the annual mean precipitation. I resample to yearly means, give the time axis a numeric coordinate and fit a line at every grid cell.

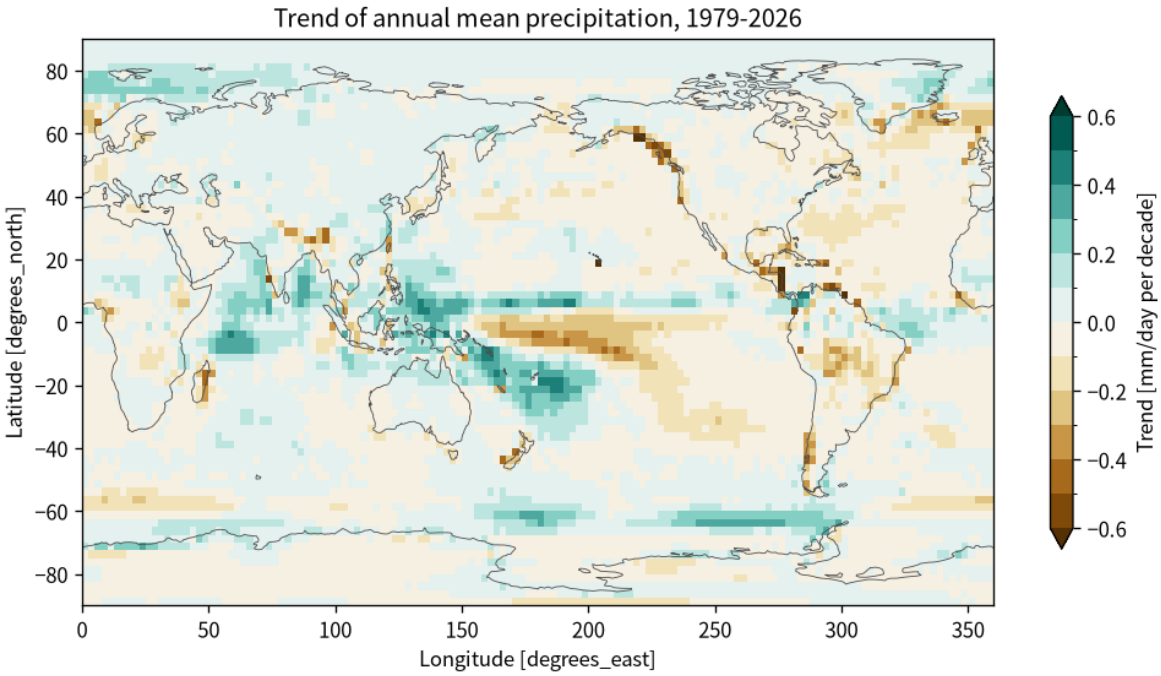
In [18]:

```
# (3) Trend of annual mean precipitation, mm/day per decade
precip_year = precip.resample(time="1YE").mean(dim="time")
t_year = precip_year.time.dt.year + 0.5

trend = precip_year.assign_coords(time=t_year).polyfit(dim="time", deg=1)
trend_slope = trend["polyfit_coefficients"].sel(degree=1) * 10

fig = plt.figure(figsize=(10, 5), dpi=120)
ax = fig.add_subplot()
trend_slope.plot(ax=ax, cmap="BrBG",
                 levels=np.arange(-0.6, 0.65, 0.1), extend="both",
                 cbar_kwargs={"label": "Trend [mm/day per decade]", "shrink": 0.8})
draw_coast(ax)
ax.set_title("Trend of annual mean precipitation, 1979-2026")

plt.savefig("figs/fig3_2c.png", bbox_inches="tight")
plt.show()
```



The trends are small, between -0.6 and +0.5 mm/day per decade, so a ±1.5 color scale would show a nearly blank map. Drying appears over the tropical South Pacific, south-eastern South America, the Mediterranean and south-west Asia, and wetting at high northern latitudes and over the western equatorial Pacific. No significance test was applied, so this is a qualitative reading.

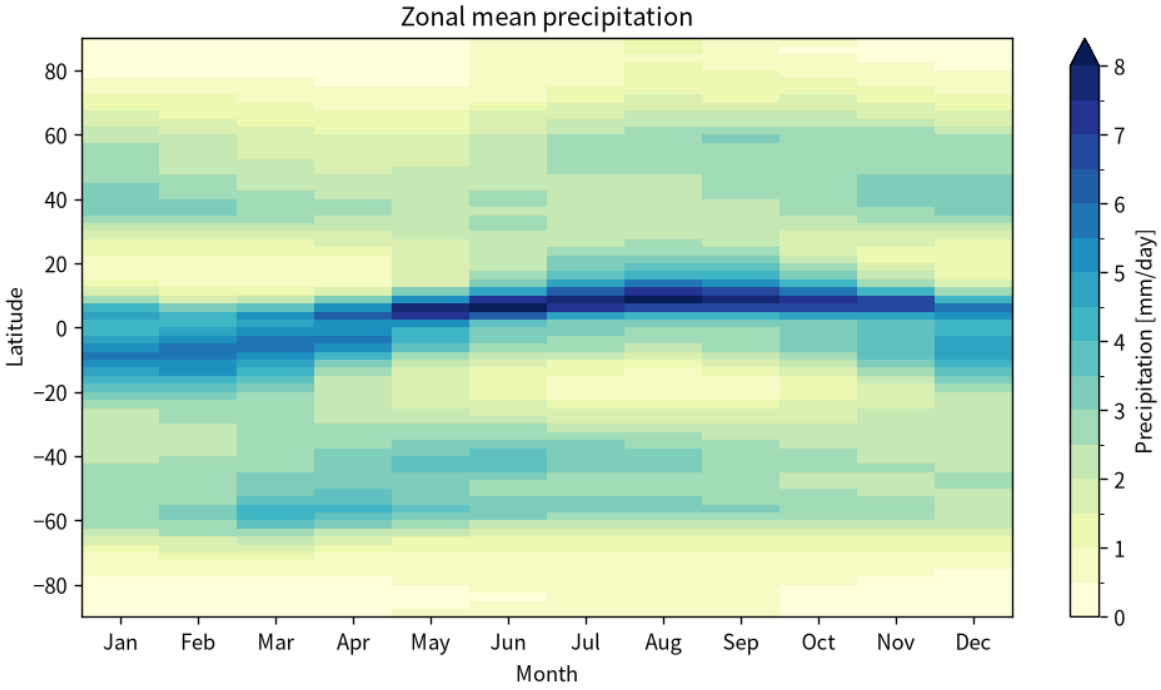
(4) The zonal mean seasonal cycle, latitude against month.

In [19]:

```
# (4) Zonal mean seasonal cycle, latitude against month
zonal = precip.mean(dim="lon").groupby("time.month").mean(dim="time")

fig, ax = plt.subplots(figsize=(10, 5), dpi=120)
zonal.plot(ax=ax, x="month", y="lat", cmap="YlGnBu", levels=np.arange(0, 8.1, 0.5),
           extend="max", cbar_kwargs={"label": "Precipitation [mm/day]"})
ax.set(xlabel="Month", ylabel="Latitude", title="Zonal mean precipitation")
ax.set_xticks(np.arange(1, 13))
ax.set_xticklabels(month_names)

plt.savefig("figs/fig3_2d.png", bbox_inches="tight")
plt.show()
```



The ITCZ sits near 5-10°S in February-March and near 10°N in July-August. The storm-track band at 40-60°S is wet all year and wetter in the local winter, and the subtropical dry belt at 20-30°N stays dry in every month.

(5) Three regions side by side: South China, the Amazon and the Sahel.

In [20]:

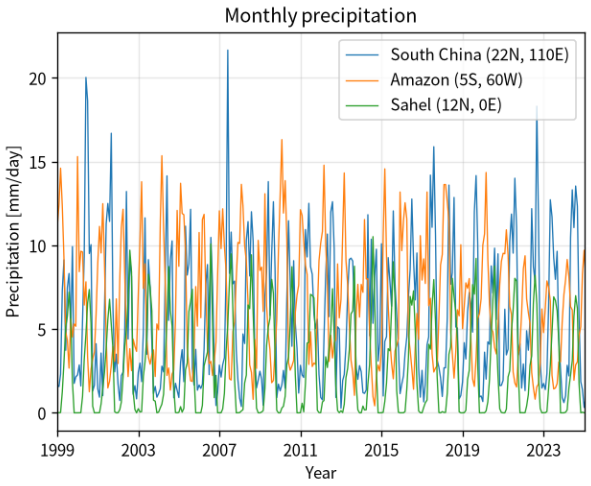
```
# (5) Three regions: monthly series and seasonal cycle
# Longitude follows the 0-360 convention of this file, so 60W is 300E
regions = {"South China (22N, 110E)": (22, 110),
           "Amazon (5S, 60W)": (-5, 300),
           "Sahel (12N, 0E)": (12, 0)}

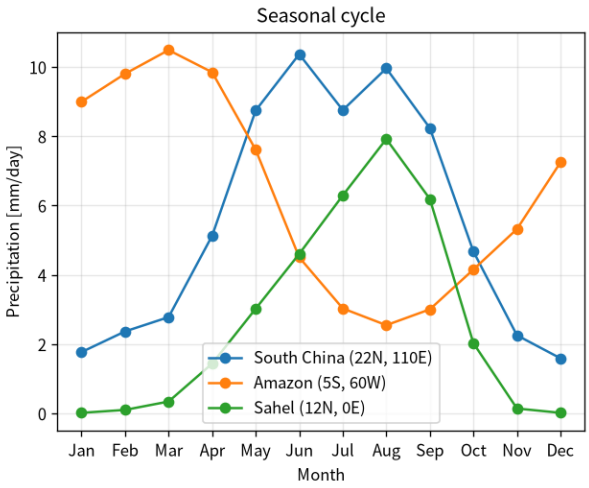
fig, axes = plt.subplots(1, 2, figsize=(13, 4.5), dpi=120)

for name, (lat, lon) in regions.items():
    series = precip.sel(lat=lat, lon=lon, method="nearest")
    axes[0].plot(series.time, series, linewidth=0.8, label=name)
    axes[1].plot(np.arange(1, 13), series.groupby("time.month").mean(dim="time"),
                 marker="o", label=name)

axes[0].set(xlabel="Year", ylabel="Precipitation [mm/day]", title="Monthly precipitation")
axes[0].set_xlim(np.datetime64("2000-01-01"), np.datetime64("2026-01-01"))
axes[1].set(xlabel="Month", ylabel="Precipitation [mm/day]", title="Seasonal cycle")
axes[1].set_xticks(np.arange(1, 13))
axes[1].set_xticklabels(month_names)
for ax in axes:
    ax.grid(alpha=0.3)
    ax.legend()

plt.savefig("figs/fig3_2e.png", bbox_inches="tight")
plt.show()
```





The three annual cycles have completely different phases. South China peaks in June with about 10 mm/day and drops below 2 mm/day in winter, the Amazon peaks in March-April, and the Sahel has a single peak above 6 mm/day in July-August and is almost dry for the rest of the year. The left panel also shows how large the year-to-year swings are, for example the Sahel drought of the 1980s and its recovery after the 1990s, although one 2.5° cell is not the whole region.

(6) The distribution of all monthly values, and the same values grouped by latitude band.

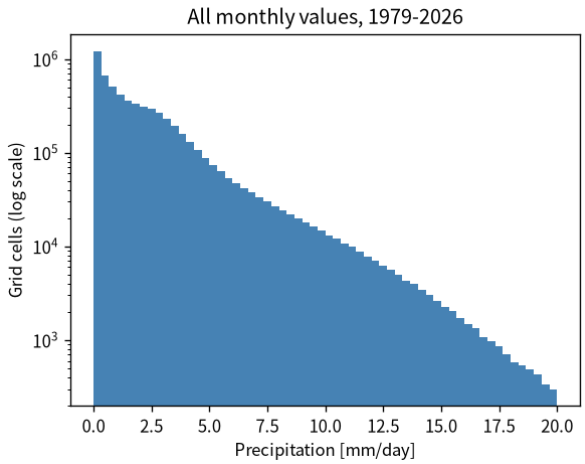
In [21]:

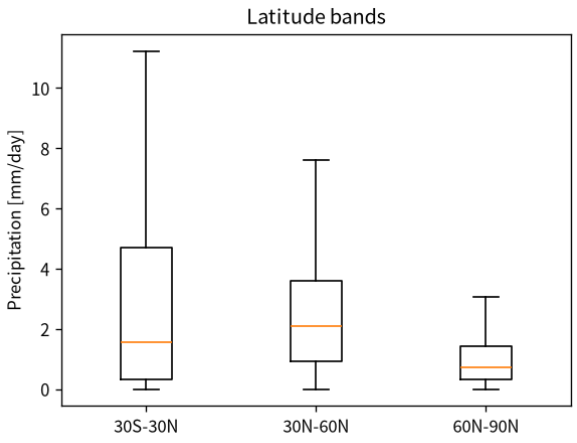
```
# (6) Distribution of the monthly values
fig, axes = plt.subplots(1, 2, figsize=(12, 4), dpi=120)

values = precip.values[~np.isnan(precip.values)]
axes[0].hist(values, bins=60, range=(0, 20), color="steelblue")
axes[0].set_yscale("log")
axes[0].set(xlabel="Precipitation [mm/day]", ylabel="Grid cells (log scale)",
            title="All monthly values, 1979-2026")

# Latitude bands
axes[1].boxplot([precip.sel(lat=slice(-30, 30)).values.ravel(),
                 precip.sel(lat=slice(30, 60)).values.ravel(),
                 precip.sel(lat=slice(60, 90)).values.ravel()],
                tick_labels=["30S-30N", "30N-60N", "60N-90N"], showfliers=False)
axes[1].set(ylabel="Precipitation [mm/day]", title="Latitude bands")

plt.savefig("figs/fig3_2f.png", bbox_inches="tight")
plt.show()
```





The distribution is strongly skewed: more than half of the cells have less than 1 mm/day and the tail reaches past 20 mm/day, so the histogram needs a log scale. By latitude the tropics (30°S-30°N) have a median near 1.6 mm/day and the widest spread, 30°N-60°N a slightly higher median (2.1 mm/day) with a narrower range, and 60°N-90°N less than 1 mm/day with almost no tail. Convective rain dominates in the tropics, stratus and light rain at high latitudes.

Notes

The methane and ERSST files were provided with the assignment; the GPCP file comes from the NOAA PSL mirror (psl.noaa.gov/data/gridded/data.gpcp.html) and is stored locally, so the notebook runs offline. I checked the methane curve against the NASA SVS animation and the Niño 3.4 figure against the NOAA monitoring page. The notebook and this report were written by myself, and the plotting follows the Section 07 and Section 08 notebooks.